AP PRECALCULUS

Unit 1 • Lesson 1.11 notes

Lesson 1.11 • Equivalent Representations of Polynomial and Rational Expressions

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Rewrite polynomials in standard and factored form.
- Divide polynomials (long and synthetic division) and use the remainder theorem.
- Expand powers of binomials with Pascal's triangle.

Vocabulary. standard form, factored form, quotient, remainder, synthetic division, binomial theorem.

1. Each form shows something

Standard form shows the degree, leading coefficient and y-intercept; factored form shows the zeros; quotient-plus-remainder form shows end behavior and slant asymptotes of rational functions.

2. Division and the remainder theorem

Key idea

Dividing $p(x)$ by $(x-a)$ gives $p(x) = (x-a)q(x) + R$, where the remainder $R = p(a)$. So $(x-a)$ is a factor exactly when $p(a) = 0$.

Example 1 • Synthetic division

Divide $2x^3 - 3x^2 - 11x + 6$ by $x - 3$.

3	2	-3	-11	6
		6	9	-6
	2	3	-2	0

Solution. Quotient $2x^2 + 3x - 2 = (2x-1)(x+2)$, remainder 0. So $2x^3 - 3x^2 - 11x + 6 = (x-3)(2x-1)(x+2)$, with zeros $3,\ \frac{1}{2},\ -2$.

3. Binomial theorem

The coefficients of $(a+b)^n$ are row n of Pascal's triangle:

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 1 & & 1 & & \\ & 1 & & 2 & & 1 & \\ 1 & & 3 & & 3 & & 1 \\ & 1 & & 4 & & 6 & & 4 & & 1 \\ 1 & & 5 & & 10 & & 10 & & 5 & & 1 \end{array}$$

Example 2

Expand $(x-2)^4$.

Solution. $x^4 + 4x^3(-2) + 6x^2(-2)^2 + 4x(-2)^3 + (-2)^4 = x^4 - 8x^3 + 24x^2 - 32x + 16$.

Common mistakes

- Leaving out a 0 coefficient for a missing power in synthetic division (for x^3+5, use $1, 0, 0, 5$).
- Dropping the signs inside $(x-2)$ when expanding: every odd power of -2 is negative.

Quick check

1. Remainder when $x^4 - 3x + 1$ is divided by $x + 1$?
2. Coefficient of x^2 in $(x+3)^3$?

Answers. 1. $p(-1) = 1+3+1 = 5$. 2. $3 \cdot 3 = 9$.